Phishing Email Detection & Awareness System

Professional Security Analysis Report

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Course: Cyber Security Fundamentals

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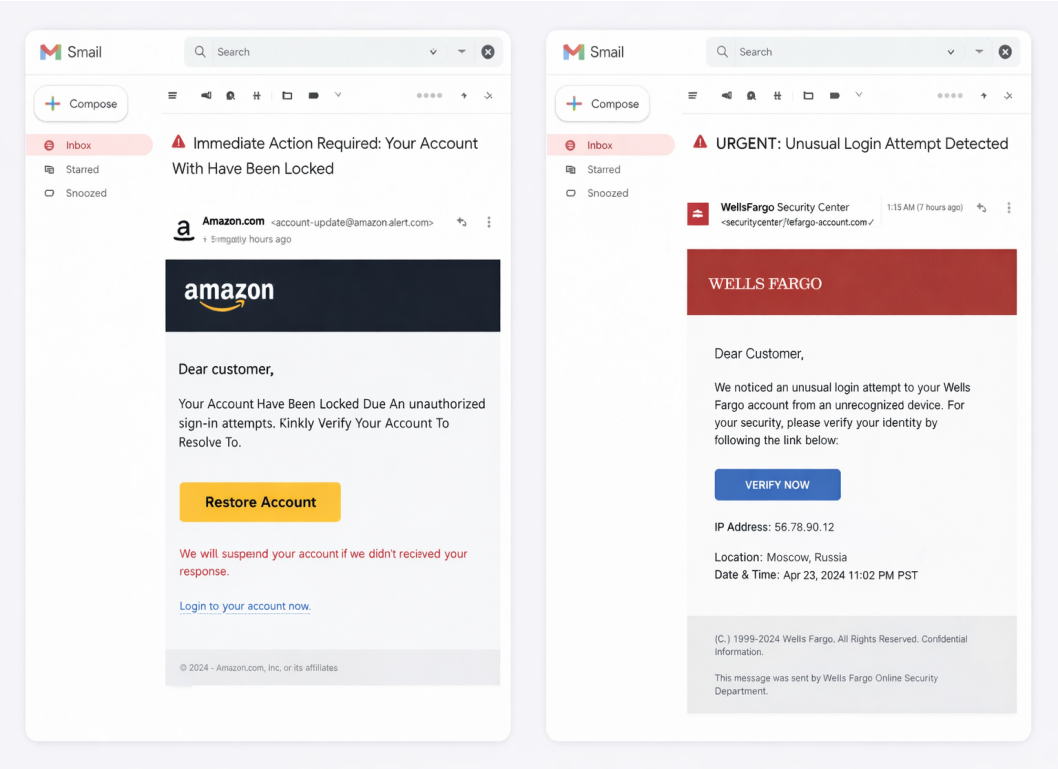
1. Executive Summary

This report presents a structured analysis of phishing email samples and one legitimate email sample. The objective is to identify phishing indicators, evaluate associated risks, and provide preventive security awareness guidelines. Phishing remains one of the most common cyber threats targeting individuals and organizations through social engineering techniques.

2. Methodology

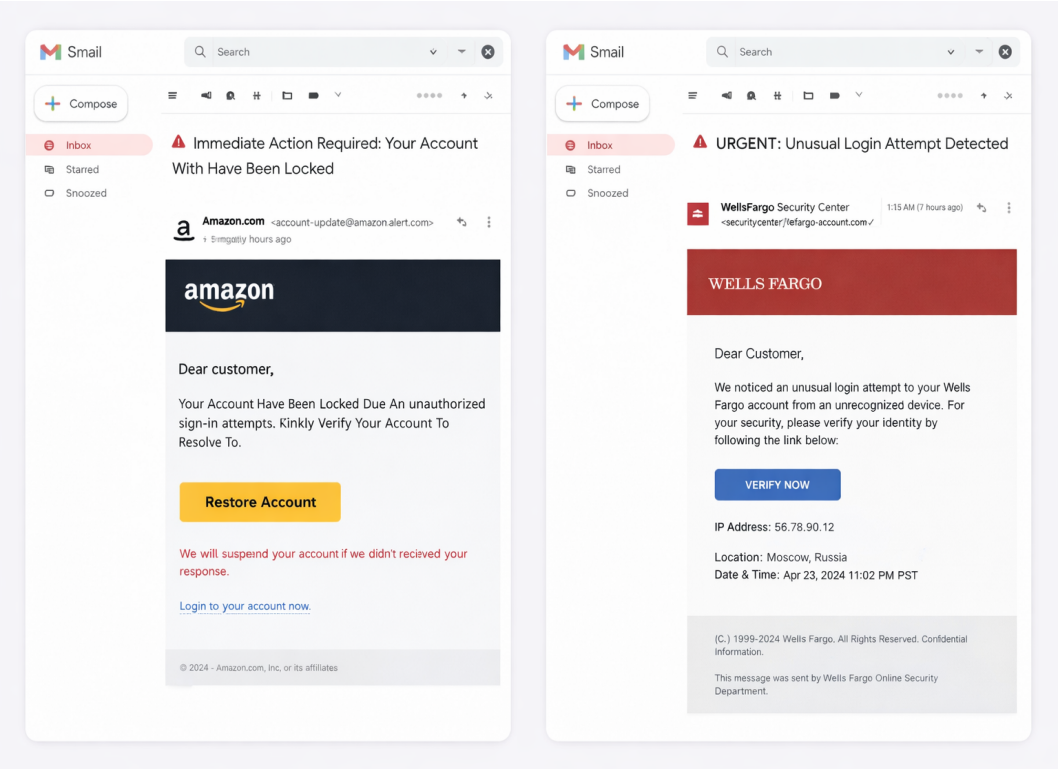
- Collected sample phishing and legitimate emails for analysis.
- Examined sender domain authenticity.
- Reviewed subject line urgency and tone.
- Analyzed grammar, structure, and content manipulation.
- Inspected embedded links and call-to-action buttons.
- Classified emails as Safe or Phishing based on findings.

3. Email Analysis – Phishing Sample 1 (Amazon)



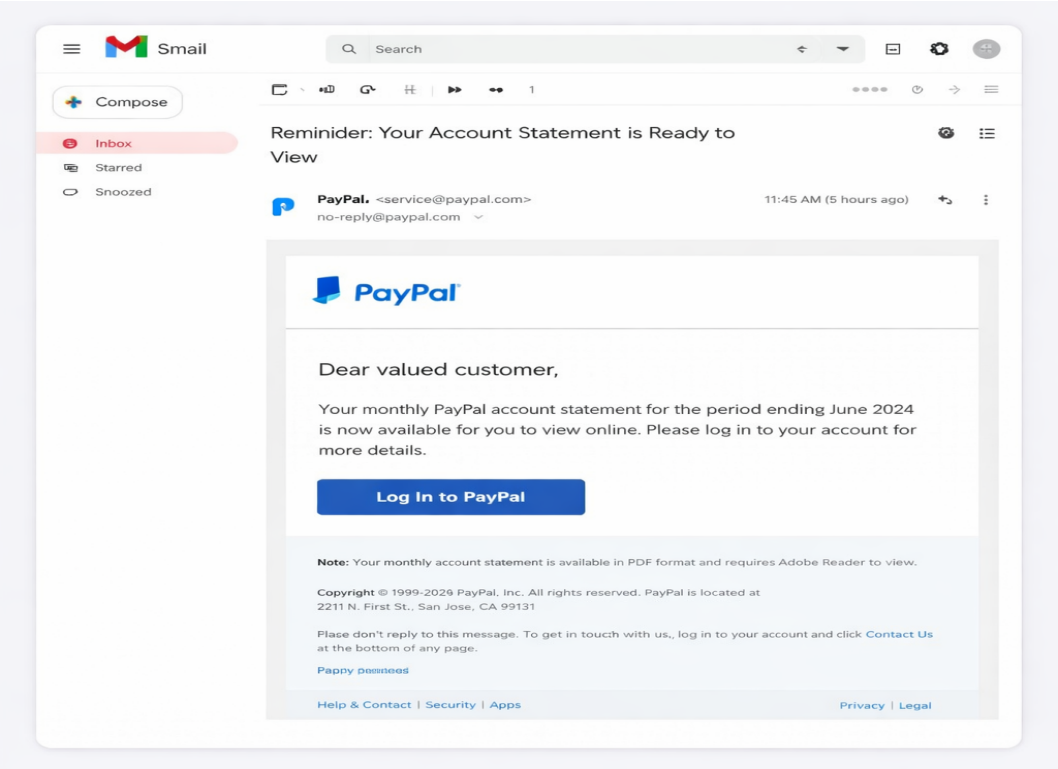
- Spoofed sender domain (amazon.alert.com).
- High urgency subject line creating panic.
- Multiple grammatical errors indicating fraud.
- Threatening suspension warning.
- Suspicious call-to-action button.

4. Email Analysis – Phishing Sample 2 (Wells Fargo)



- Fake domain (wellsfargo-account.com).
- Unusual login attempt alert to trigger fear.
- Requests identity verification via link.
- Uses geographic/IP scare tactics.
- Encourages immediate action without verification.

5. Email Analysis – Legitimate Sample (PayPal)



- Verified official domain (paypal.com).
- Professional and consistent branding.
- No excessive urgency or threats.
- Clear contact and support information.
- Standard account statement notification.

6. Risk Impact Assessment

- Credential theft leading to account compromise.
- Financial fraud and unauthorized transactions.
- Identity theft and data exposure.
- Malware infection through malicious links.
- Reputation damage for organizations.

7. Comparative Analysis

Feature	Legitimate Email	Phishing Email
Sender Domain	Official domain	Misspelled/Fake domain
Tone	Professional	Urgent/Threatening
Grammar	Correct	Errors present
Links	Official website	Suspicious link
Intent	Informational	Data theft

8. Prevention & Security Awareness Guidelines

- Verify sender email addresses before taking action.
- Hover over links to inspect actual URL destinations.
- Enable Multi-Factor Authentication (MFA).
- Avoid clicking links from unknown or suspicious emails.
- Report phishing attempts immediately.
- Educate users regularly on social engineering tactics.

9. Conclusion

Phishing attacks rely primarily on social engineering techniques rather than technical vulnerabilities. Through structured analysis, this report identified clear indicators such as spoofed domains, urgency tactics, and malicious links. Strengthening user awareness and implementing security controls significantly reduces the risk of successful phishing attacks.